

HOLBEACH, United KingdomWMO: **034690**Lat: **52.8733N**Lon: **0.1386E**Elev: **10**StdP: **14.69**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.1	29.3	23.2	17.5	31.2	25.5	19.5	32.1	35.8	48.2	32.4	48.5	8.6	320	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	78.0	66.3	74.7	64.7	72.1	63.3	68.2	75.0	66.3	72.5	64.7	70.3	12.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.4	71.9	63.9	88.8	69.5	62.3	83.9	67.7	32.4	75.1	30.9	72.6	29.7	70.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.4	27.1	24.3	21.9	86.4	4.6	3.2	18.6	88.7	15.9	90.6	13.3	92.4	9.9	94.7		
(5)			21.9	73.2	4.7	3.7	18.5	75.9	15.8	78.1	13.1	80.2	9.7	82.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	40.1	40.9	43.8	47.9	53.1	58.2	62.2	62.2	57.9	52.4	45.1	40.8
	DBStd	9.48	5.25	5.55	5.12	5.13	4.91	4.60	4.42	4.33	4.29	5.21	5.40	6.09
	HDD50	1379	308	259	201	100	24	1	0	0	1	34	161	290
	HDD65	5377	772	676	657	514	370	210	111	110	218	392	596	751
	CDD50	1534	1	3	9	35	120	247	377	378	237	108	15	4
	CDD65	57	0	0	0	0	0	7	23	23	3	1	0	0
	CDH74	372	0	0	0	1	8	63	145	128	23	4	0	0
	CDH80	55	0	0	0	0	0	9	22	23	1	0	0	0
Wind	WSAvg	12.5	13.9	13.8	13.5	12.9	12.8	11.3	10.5	10.8	11.7	12.6	12.6	13.5
Precipitation	PrecAvg	23.9	2.0	1.6	1.4	1.7	1.9	2.1	2.2	2.4	1.9	2.4	2.3	2.0
	PrecMax	29.5	3.6	2.7	2.7	4.7	4.4	5.5	4.6	4.4	3.7	3.9	4.3	3.9
	PrecMin	16.1	0.6	0.3	0.2	0.1	0.6	0.4	0.6	0.3	0.3	0.7	1.1	0.9
	PrecStd	4.1	0.9	0.6	0.7	1.3	0.9	1.3	1.1	1.1	1.0	0.9	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.4	61.2	69.4	74.0	80.9	82.4	83.1	77.2	70.0	60.4	56.7
		MCWB	52.4	53.1	53.7	58.3	62.1	67.5	69.4	69.0	64.4	62.5	57.6	53.7
	2%	DB	52.6	53.9	57.7	63.9	69.2	74.6	77.8	77.2	72.4	65.0	57.7	54.4
		MCWB	50.0	50.3	52.0	55.4	60.5	64.6	66.3	66.7	63.0	59.6	55.1	52.1
	5%	DB	50.4	51.6	55.1	60.1	65.4	70.7	74.5	73.9	68.5	62.4	55.5	52.2
		MCWB	48.2	48.2	50.3	52.9	57.9	62.3	64.8	64.2	61.0	58.3	53.0	50.1
	10%	DB	48.2	49.4	52.6	57.1	62.3	67.5	71.5	71.1	65.9	60.1	53.3	50.0
		MCWB	46.1	46.7	48.5	51.2	55.7	60.1	63.5	63.0	59.5	56.6	51.2	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	53.8	55.3	60.0	65.1	69.7	71.3	70.9	66.7	63.2	58.2	54.4
		MCDB	54.7	57.0	59.5	67.1	70.7	78.8	80.0	79.0	73.3	68.4	60.1	56.0
	2%	WB	50.5	50.7	52.9	56.6	61.4	65.5	68.2	68.0	64.3	60.9	55.5	52.5
		MCDB	52.2	53.5	56.6	62.3	68.0	72.0	74.9	74.7	70.2	64.3	57.4	54.3
	5%	WB	48.5	48.8	50.9	54.0	58.9	63.2	66.2	66.2	62.3	59.0	53.3	50.3
		MCDB	50.3	51.1	54.3	58.9	63.7	68.9	72.5	72.1	66.9	61.6	55.1	52.0
	10%	WB	46.4	47.0	49.0	51.8	56.8	61.2	64.5	64.4	60.7	57.2	51.5	48.2
		MCDB	48.0	49.3	52.2	56.3	61.2	66.3	70.2	69.3	64.5	59.7	53.1	49.9
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.6	12.9	14.7	14.8	15.1	16.0	16.2	14.9	12.4	10.4	8.6
		MCDBR	11.0	12.5	16.5	20.0	20.3	20.7	21.0	20.3	19.0	13.9	11.2	10.4
		MCWBR	10.3	10.0	11.6	12.5	12.7	12.2	11.4	10.9	11.2	9.6	9.6	9.7
	5% WB	MCDBR	11.0	12.3	15.5	18.6	18.1	18.9	19.0	18.1	16.5	13.0	11.1	10.3
		MCWBR	10.4	10.3	11.6	12.5	12.4	12.3	11.6	11.1	10.9	10.0	9.9	9.8
Clear-Sky Solar Irradiance	taub	0.326	0.343	0.376	0.402	0.403	0.409	0.410	0.397	0.376	0.356	0.329	0.315	
	taud	2.358	2.382	2.286	2.227	2.271	2.277	2.287	2.342	2.400	2.437	2.407	2.338	
	Ebn at Noon	207	242	255	263	269	268	265	262	253	233	205	187	
	Edh at Noon	19	25	33	40	40	40	39	35	30	23	18	16	
All-Sky Solar Radiation	RadAvg	223	441	800	1250	1545	1624	1564	1311	969	556	278	178	
	RadStd	21	60	86	132	112	133	156	90	83	46	29	20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page